Istanbul Technical University - Computer Engineering
BLG 561E - Deep Learning
Homework 3
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Oh baby!
That's why I'm easy
I'm easy like Sunday morning

Easy, Lionel Richie

Part 1 (30 pts) - StyleGAN
In this part of the homework, we will use StyleGANv3[1] to generate realistic face images. To do this, we will use the pretrained network with a simple script as given in the following code.

```
import pickle
import matplotlib.pyplot as plt
import torch
import cv2
import numpy as np
with open("stylegan3-t-ffhq-1024x1024.pkl", "rb") as f:
  a = pickle.load(f)
#Unpack the GAN network downloaded from the official NVIDIA website.

gan = a["G_ema"]
#StyleGAN contains both the Generator and the Discriminator. We only
    need the Generator to generate the images.
gan.eval()
#In PyTorch, every network has to be set as 'train' or 'eval'. To
    evaluate an input via the network, we should set it as eval.
for param in gan.parameters():
  param.requires_grad = False
#When training the network, the parameter change is done via the
    gradients. Setting that none of the parameters need gradients will
     speed up our process.

z = torch.randn(1, 512)
# The StyleGAN generator takes a vector of size (512) to generate an
    image.
img = gan(z, 0).numpy().squeeze()
#Obtain the image from the generator.

img = np.transpose(img, (1,2,0))
```

[1]Karras, Tero, et al. "Alias-free generative adversarial networks." arXiv preprint arXiv:2106.12423 (2021)

```
#Initially, the output of the network is like (RGB Channel, Width,
    Height). We should change the order of the channels to make it
    look like an OpenCV image.

img[img>1] = 1
img[img<-1] = -1
img = 255*(img + 1) / 2
#The network outputs images with float values, centered at zero. Thus
    , we should make the values between [0-255].

cv2.imwrite('test.png', img[:,:,[2,1,0]])
```

Create two random face images using StyleGAN generator. To do this, you should download *stylegan3-t-ffhq-1024x1024.pkl* from NVIDIA's official page[2]. The model here creates 1024x1024 sized images. If your system is not sufficient, you can also use other models from the same site for 256x256 outputs.

Figure 1: Two face images created using StyleGAN.

Part 2 (30 pts) - StyleGAN Animation

In the previous part, you have created two face images using two 512 sized vectors. Here we will create an animation of creating the second face by changing the first face. To do this, start by creating the first face and then slowly changing the face vector step-by-step by a small amount, obtain the second face. Some examples can be found under my website[3].

[2] https://catalog.ngc.nvidia.com/orgs/nvidia/teams/research/models/stylegan3/files

[3] `https://web.itu.edu.tr/sahinyu/face1_video.mp4`, `https://web.itu.edu.tr/sahinyu/face2_video.mp4`, `https://web.itu.edu.tr/sahinyu/face3_video.mp4`

Part 3 (40 pts) - Pixel2Style2Pixel

Pixel2Style2Pixel Framework [4] creates vectors representing the given face. The procedure could be seen in Figure 2.

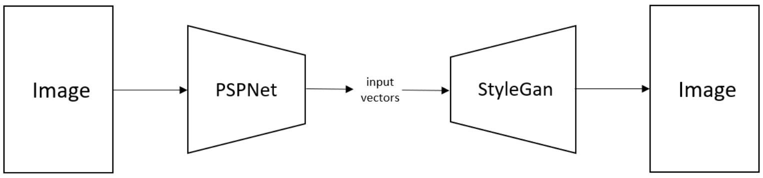

Figure 2: PSPNet and StyleGAN.

Create the animation as in Part 2, using images from your favourite actors or famous people! Some examples could be found in my website [5].

Especially for this part, it is suggested to use Google Colab. First of all, download the source code I prepared by simplifying the original PSPNet GitHub. You could find the code at the following link [6].

[4] Richardson, Elad, et al. "Encoding in style: a stylegan encoder for image-to-image translation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2021.

[5] https://web.itu.edu.tr/sahinyu/hw4/bruce.mp4,https://web.itu.edu.tr/sahinyu/hw3/leo.mp4,https://web.itu.edu.tr/sahinyu/hw3/nbc.mp4

[6] https://web.itu.edu.tr/sahinyu/hw3/PSPNet_simplified.zip